

Mathematical Framework for HVK

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1 Introduction

Classical Neural Networks exploit spatial and translational symmetry throughout learned filter kernels. Quantum Machine Learning(QML) offers a complementary inductive bias : the Hilbert space structure of a multi qubit systems provides exponential rich feature space , while the entanglement encodes non-local correlations which are computationally expensive to represent classically

The Hamiltonian Vision Kernel(HVK) connects these two paradigms. At its core the HVK treats each image patch as a quantum-many-body wave function compresses it via a Matrix Product State(MPS) with finite bond dimension , and reads out the Pauli expectation values. These features are subsequently processed by a Variational Quantum Circuit (VQC) with parameterized learnable weights whose energy is minimized using a spatial-war Heisenberg Hamiltonian. Classical MLP decoder maps 27 dimensional observable vector to pixel space. The remainder of this particular document is broken into 7 parts which explicitly describes the MPS extraction , Fourier positional encoding , the variational circuit and the construction of SU(3) variant latent space , positional aware Heisenberg Hamiltonian loss function , the symmetry structure of the system and tracing the transformation of symmetry from SU(3) to D4

2 System Specifications

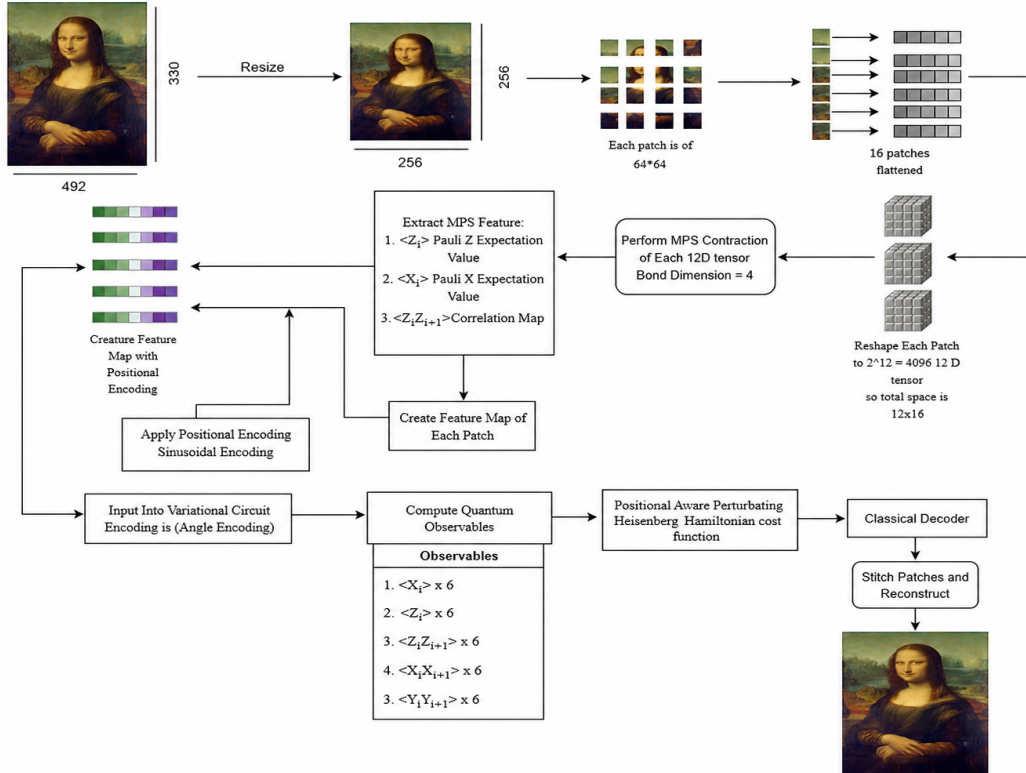


Figure 1: Full architecture of HVK

The above figure essentially shows the system architecture of the Hamiltonian Vision Kernel , showing the most important features of the architectures including the :

- Fourier Positional Encoding
- Latent Vector Space and Creation of SU(3) variant Latent space of Observable expectation Values
- Perturbation Hamiltonian Loss Function and the Calculated Correlations of pixel to pixel Observables

we will discuss in details of each and every component of the HVK making it mathematically rigorous and also discuss about the symmetry transformation how the systems changes the SU(3) symmetry group and makes it D4 dihedral group for high image resolution and Ultimately this change in the symmetry group essentially makes the architecture interesting mathematically.

3 Matrix Product State Encoding and Feature Extraction

3.1 Patch Preparation

An input image $I \in \mathbb{R}^{H \times W}$ is resized to a fixed resolution of 256×256 pixels and normalised to the range $[0, 1]$. The image is partitioned into a regular grid of non-overlapping 64×64 patches:

$$\{P_{ij} \in \mathbb{R}^{64 \times 64} : i, j \in \{0, 1, 2, 3\}\},$$

yielding a total of

$$N = 16$$

patches Each patch is associated with a normalized grid coordinate given by the mathematical expression

$$(r_i, c_j) = \left(\frac{i}{4}, \frac{j}{4}\right) \in [0, 1]^2$$

3.2 Mps Feature Extraction

Each 64×64 patch P is vectorized to obtain

$$v \in \mathbb{R}^{4096},$$

which is L_2 -normalised and reshaped into a rank-12 tensor state:

$$\psi = \frac{v}{\|v\|_2} \in (\mathbb{C}^2)^{\otimes 12}.$$

treating each of the 12 binary indices in the physical site of then quantum spin-1/2 chain (nsites = 12 and physical dimension $d = 2$). This interpretation is exact since 2^{12} is 4096.

The normalised state $|\psi\rangle$ is then compressed into a Matrix Product State (MPS) with bond dimension $\chi = 4$:

$$|\psi\rangle \approx \sum_{\sigma_1, \dots, \sigma_{12}} A_1^{\sigma_1} A_2^{\sigma_2} \dots A_{12}^{\sigma_{12}} |\sigma_1 \dots \sigma_{12}\rangle,$$

where each tensor satisfies

$$A_i^{\sigma_i} \in \mathbb{C}^{\chi_{i-1} \times \chi_i}, \quad \chi_i \leq \chi = 4.$$

The compression is performed via sequential singular value decomposition (SVD), truncating singular values beyond rank χ at each bond.

The compressed MPS is normalised such that

$$\langle \psi | \psi \rangle = 1,$$

ensuring that expectation values are well-defined.

3.3 Observable Feature Extraction

Three families of observables are calculated from the MPS , collectively constituting the quantum latent feature vector $f \in \mathbb{R}^{46}$ for each patch cause there are 12 features so total of 46 vector space.

3.3.1 Local Pauli Expectations value of the System

for each site $i \in 0, \dots, 11$, the expectation values of Pauli Z and Pauli X operators are calculated :

$$\langle Z_i \rangle = \langle \psi | I^{\otimes(i-1)} \otimes Z \otimes I^{\otimes(11-i)} | \psi \rangle,$$

$$\langle X_i \rangle = \langle \psi | I^{\otimes(i-1)} \otimes X \otimes I^{\otimes(11-i)} | \psi \rangle,$$

where

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

are the standard 2×2 Pauli matrices.

This yields

$$2 \times 12 = 24$$

features.

3.3.2 Nearest Neighbor Correlations

For each bond (i , i+1) the two site ZZ correlator is computed:

$$\langle Z_i Z_{i+1} \rangle = \langle \psi | Z_i \otimes Z_{i+1} | \psi \rangle$$

computed efficiently using by applying Z gate at site i and i+1 to a copy of the MPS. This contributes to 11 features.

3.3.3 Bipartite Entanglement Entropy

At each of the 11 internal bonds $i \in 1, \dots, 11$, the bipartite entanglement entropy S_i is computed using from The Schmidt spectrum $\{\lambda_k^{(i)}\}$ is obtained directly from the MPS representation:

$$p_k^{(i)} = \frac{\left(\lambda_k^{(i)}\right)^2}{\sum_k \left(\lambda_k^{(i)}\right)^2},$$

$$S_i = - \sum_k p_k^{(i)} \log(p_k^{(i)} + \varepsilon),$$

where

$$\varepsilon = 10^{-8}$$

prevents numerical divergence. This contributes 11 features.

The complete feature vector is given by

$$f = (\langle Z_0 \rangle, \langle X_0 \rangle, \dots, \langle Z_{11} \rangle, \langle X_{11} \rangle, \langle Z_0 Z_1 \rangle, \dots, \langle Z_{10} Z_{11} \rangle, S_1, \dots, S_{11}) \in \mathbb{R}^{46}.$$

The feature matrix

$$F = \{f_n\}_{n=1}^{16} \in \mathbb{R}^{16 \times 46}$$

is column-normalised (zero mean and unit variance per feature dimension) to standardise the circuit input scale.

Let us formulate bipartite entanglement more rigorously. Consider an arbitrary matrix

$$M \in \mathbb{C}^{m \times n}.$$

The singular value decomposition of M is given by the formula

$$M = U \Sigma V^\dagger,$$

where

$$U \in \mathbb{C}^{m \times m}, \quad V \in \mathbb{C}^{n \times n}$$

are unitary matrices, and

$$\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r),$$

with

$$\sigma_i \geq 0, \quad r = \text{rank}(M),$$

denotes the singular value matrix.

Equivalently,

$$M = \sum_{i=1}^r \sigma_i |u_i\rangle\langle v_i|,$$

where $\{|u_i\rangle\}$ and $\{|v_i\rangle\}$ are orthonormal left and right singular vectors respectively.

If the matrix M represents the coefficient tensor of a bipartite quantum state

$$|\psi\rangle = \sum_{a=1}^m \sum_{b=1}^n M_{ab} |a\rangle \otimes |b\rangle,$$

then the singular values $\{\sigma_i\}$ correspond exactly to the Schmidt coefficients of the bipartite decomposition:

$$|\psi\rangle = \sum_{i=1}^r \sigma_i |u_i\rangle \otimes |v_i\rangle.$$

The normalized Schmidt probabilities are therefore given by

$$p_i = \frac{\sigma_i^2}{\sum_{j=1}^r \sigma_j^2}.$$

The bipartite entanglement entropy is then defined as the von Neumann entropy

$$S = - \sum_{i=1}^r p_i \log p_i.$$

For numerical stability, one may introduce a small regularization parameter $\epsilon > 0$:

$$S_\epsilon = - \sum_{i=1}^r p_i \log(p_i + \epsilon).$$

At this stage, the numerical procedure itself is relatively straightforward. However, a deeper and more important question naturally arises: *Why are the feature observables computed specifically from the MPS-compressed wavefunctions, and why are only certain observables selected?* More precisely, one may ask why the architecture primarily evaluates expectation values of local operators such as

$$\langle X_i \rangle, \quad \langle Z_i \rangle,$$

instead of considering more general two-body correlators such as

$$\langle X_i X_{i+1} \rangle, \quad \langle Y_i Y_{i+1} \rangle,$$

or even higher-order non-local observables. These questions are both valid and fundamentally important, since they probe the underlying symmetry structure and information flow encoded within the architecture itself. The choice of observables are not arbitrary they are linked with the architecture. When you compress a patch into an MPS and read the extracted feature, as MPS framework being totally built on

the computation $\langle Z \rangle$ basis. The Schmidt decomposition that the QUIMB library of python is performing is performing entirely aligned with the Z axis bond. So the ZZ corelations are the most immediate consequential inner product of the MPS structure. Now if we had considered the XX or YY other corelators what we had to change the MPS basis by rotating the Z axis using a Hadamard for X and Hadamard+S Gate for the Y which lead to the increase of both bond dimensional and the computational cost of the system. Moreover the rotated basis would eventually the capture the same underlying entanglement providing redundant information which is not required to describe the physics of the system. This was the mathematical and technical description why XX and YY are not used for understanding thr underlying entanglement. Now we discuss about the symmetry perspective of the system now. So systems after using the MPS framework the systems becomes symmetric under Z and is asymmetric on the X and Y so this is essentially this enables the spatially aware Heisenberg Loss Function tries to minimize. So if we try to enforce all the XX , YY , ZZ symmetries on the system this makes the feature space $SU(3)$ symmetric so essentially leads to a balanced isotropic Heisenberg Hamiltonian essentially making the entire architecture invariant under $SU(3)$ unable to make the system spatially aware resulting in patch overlapping and decreases in quantitative metrics.

now finally we generate the feature space as

$$F = f_{n=1}^{16} \in \mathbb{R}^{16 \times 46}$$

which is column normalised meaning zero mean , and unity variance to standardize the circuit input scale.

4 Positional Encoding

4.1 Background

The MPS feature vector $\mathbb{F}_n$ encodes only the spatial content of the patch $\mathbb{P}_n$ and is by construction permutation invariant with respect to the patch ordering precisely. Now to make the system spatially discriminative in nature i.e different patches produce different outputs from the variational circuit each patch is augmented with a position-aware encoding derived from the grid coordinates.

4.2 Two-Dimensional Sinusoidal encoding

The positional encoding for this hybrid transformer architecture is based purely on the attention mechanism is based on the architecture given by Vaswani et.al (2017) [1] and is adapted for 2D spatial coordinate system , we define a encoded vector space $\mathbb{D}_{enc} = 8$ dimensional Fourier positional encoding. For a patch normalized grid coordinate $(r_i, c_j) \in [0, 1]^2$, the encoded $e_{ij} \in \mathbb{R}^8$ is :

$$\begin{aligned} e_{ij}^{4K} &= \sin(\omega_k \pi r_i) \text{ where } k = 0, 1 \\ e_{ij}^{4K+1} &= \cos(\omega_k \pi r_i) \text{ where } k = 0, 1 \\ e_{ij}^{4K+2} &= \sin(\omega_k \pi c_j) \text{ where } k = 0, 1 \\ e_{ij}^{4K+3} &= \cos(\omega_k \pi c_j) \text{ where } k = 0, 1 \end{aligned}$$

where the frequency $\omega_k = \frac{1}{1000 \frac{4k}{D \times K}}$. Explicitly , the encoding matrix $\mathbb{E} \in \mathbb{R}^{16 \times 8}$ covers all the four frequency components : $\sin(r)$, $\cos(r)$, $\sin(c)$, $\cos(c)$ at two different spatial frequencies.

4.3 Learned Positional Projection

The 8 dimensional encoding is projected into $N_{qubits} = 6$ angular values via a learned linear map $W_{pos} \in \mathbb{R}^{6 \times 8}$:

$$\theta_{position} = W_{position} \cdot e \in \mathbb{R}^6$$

which serves as the per qubit RY angles in the variational circuit . The map $W_{position}$ is jointly trained along with the rest of model . This design guarantees that he citcuit is spatially aware i.e if the same MPS features is fed into the circuit at 2 different grid positions will produce different quantumn states hence different observable vectors.

5 The Variational Circuit and the $SU(3)$ Variant latent space

5.1 Circuit Architecture

The quantum processing unit consists of $N_{qubits} = 6$ and $n_{layers} = 2$ variational layers. For each patch n , the circuit is initialized at $|0\rangle^{\otimes 6}$ and three sequential operations that are applied :

- **Step 1 : Angle Embedding** - The project MPS feature Vector $x_n = W_{mps} \times f_n \in \mathbb{R}^6$ where $W_{mps} \in \mathbb{R}^{6 \times 64}$ is encoded via RY rotations :

$$U_{enc(x_n)} = \otimes_{i=0}^{[5]} RY(x_n^{[i]}), R\theta = e^{\frac{-i\theta Y}{2}}$$

- **Step 2 : Positional Modulation** - Each qubit receives an additional rotation controlled by learned positional angle:

$$U_{position(\theta_{position})} = \otimes_{i=0}^{[5]} RY(\theta_{position}^{[i]})$$

this operation breaks the positional invariance of the bare MPS features.

- **Step 3 : Strongly Entangling Layers** - A parametrized entangling ansatz with $L = 2$ layers acts on the full register:

$$U_{variational}(\theta) = U_{sel}^{[2]}(\theta^{[2]}) \cdot U_{sel}^{[1]}(\theta^{[1]})$$

where each $U_{sel}^{[1]}$ consists of single qubit rotations $Rot(\phi, \theta, \omega) = RX(\omega)RY(\theta)RZ(\phi)$ on every qubit followed by a CNOT entanglers. The trainable parametre tensor $\theta = \mathbb{R}^{2 \times 6 \times 3}$ is initialized by sampling from $U[0, \pi]$ to mitigate the possibility of barren plateaux.

5.2 The observable Latent Space

The output of this paramterized variational circuit is :

$$|\Psi_n\rangle = U_{variational} \cdot U_{postional}(\theta_{position,n}) \cdot U_{encoded}(x_n)|0\rangle^{\otimes 6},$$

for which the 27 real valued observables are simultaneously extracted:

$$\Phi_n = (\langle X_0 \rangle, \dots, \langle X_5 \rangle ; \langle Z_0 \rangle, \dots, \langle Z_5 \rangle ; \langle Z_0 Z_1 \rangle, \dots, \langle Z_4 Z_5 \rangle ; \langle X_0 X_1 \rangle, \dots, \langle X_4 X_5 \rangle ; \langle Y_0 Y_1 \rangle, \dots, \langle Y_4 Y_5 \rangle) \in [-1, 1]^{27}$$

All these 27 Observables are evaluated in a a single circuit execution(one Qnode per patch) , yielding 16×27 observable matrix $O = O_{n=1}^{16}$.

5.3 $SU(3)$ - Variant Structure of the latent space

We now establish the symmetric based study of the latent representation. The key observation for this is that the set of all states reachable by the circuit , for fixed weights θ , forms a manifold in the 6-qubit Hilbert Space $\mathbb{C}^{64}$.

Consider the lie algebra of $SU(2)^{\otimes n}$ generated by the pauli operators $\frac{\sigma_i}{2}$ where $i = 1, 2, 3$. The single qubit reductions of $|\Psi_n\rangle$ are the density metrics $\rho_i \in \mathbb{C}^{2 \times 2}$ which are the pure states that lie on a Bloch sphere $S^2 = SU(2)/U(1)$. Cause density matrices can be represented using this formula:

$$\rho_i = \frac{1}{2}(I + r \cdot \sigma_i), \text{ where } i = 1, 2, 3$$

and r is the cartesian position vector.

Now lets understand why i wrote that $S^2 \cong SU(2)/U(1)$ where $S^2 = S^1 \times S^1$ is a 2 sphere.

Lets derive the symmetry relation of the Bloch shere cause it is essentially for the understanding the basic principle of the HVK spatial exploitation.

Note: This part is mathematically rigorous explanation of the bloch sphere one might ignore cause when we will write a paper on this architecture we will not discuss so much in detail of the symmetry of the bloch sphere we will only go through the outcome .

let us consider a generalized normal qubit :

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \text{ such that } |\alpha|^2 + |\beta|^2 = 1$$

and $\alpha, \beta \in \mathbb{C}$ and the values of α and β can be written as :

$$\gamma_{\alpha, \beta} = a_k + \imath b_k$$

so there are total 4 parameters for alpha and beta when normalized one degree of freedom decreases so 3 parameters remain. In most cases what we do is that the wavefunction has a harmonic dependent phase so

$$|\psi\rangle \sim e^{\imath\phi} |\psi\rangle$$

so this phase symmetry $U(1)$ so removing another degree of freedom so the remaining 2 physical DOFs. Now a parameterised qubit in a spherical polar coordinate can be written as :

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{\imath\phi}\sin\frac{\theta}{2}|1\rangle$$

so θ, ϕ are the only remain real dofs therefore the bloch sphere is the S^2 a 2 sphere. The $SU(2)$ group by construction is defined as a special unitary group which is subgroup of $U(2)$ and is defined :

$$SU(2) = \{U \in \mathbb{C}^{2 \times 2} ; U^\dagger U = I ; \det U = 1\} \subset U(2)$$

this is the group acting on the qubits so the entire pure-state manifold is generated by a $SU(2)$ rotations. Now as the system contains both pure states and states which only changes the phase of the generalised qubit which corresponds to the $U(1)$ symmetry so the symmetry manifold of the bloch sphere is essentially $S^2 = SU(2)/U(1)$.

The two-site reduced density matrices are defined as

$$\rho_{i, i+1} = \text{Tr}_{\text{rest}} (|\Psi_n\rangle\langle\Psi_n|),$$

where $|\Psi_n\rangle$ denotes the full 12-site quantum state associated with the n -th image patch. The operation

$$\text{Tr}_{\text{rest}}$$

denotes the partial trace over all sites except i and $i + 1$. Physically, this removes the remaining degrees of freedom of the system and retains only the local quantum information shared between two neighbouring sites.

Since each site corresponds to a qubit with local Hilbert space

$$\mathbb{C}^2,$$

two neighbouring sites together live in the bipartite Hilbert space

$$(\mathbb{C}^2)^{\otimes 2} = \mathbb{C}^4.$$

Therefore,

$$\rho_{i, i+1} \in \mathbb{C}^{4 \times 4}.$$

The reduced density matrix is:

- Hermitian,
- positive semidefinite,
- trace normalised.

These are the defining properties of a quantum density matrix.

The two-site reduced density matrix contains all local pairwise quantum information between neighbouring pixels encoded in the tensor-network state, including:

- local correlations,
- quantum coherence,
- nearest-neighbour entanglement structure.

Expectation values of observables are computed directly from the reduced density matrix through

$$\langle O_i O_{i+1} \rangle = \text{Tr}(\rho_{i,i+1}(O_i \otimes O_{i+1})).$$

In the HVK architecture, the physically relevant observables are chosen as

$$\{\langle X_i X_{i+1} \rangle, \langle Y_i Y_{i+1} \rangle, \langle Z_i Z_{i+1} \rangle\},$$

which measure nearest-neighbour correlations along the three Pauli directions.

The complete set of traceless Hermitian operators acting on a two-qubit system forms the Lie algebra

$$\mathfrak{su}(4),$$

which has dimension

$$4^2 - 1 = 15.$$

This means that a general two-qubit system admits 15 independent observable generators. However, the HVK architecture does not utilise the full operator algebra. Instead, it restricts the observable sector to the symmetric nearest-neighbour correlators

$$XX, YY, ZZ,$$

which form a physically interpretable correlator manifold.

These operators behave analogously to collective spin operators and generate a reduced symmetry structure similar to an embedded

$$\mathfrak{su}(2) \subset \mathfrak{su}(4).$$

Thus, rather than learning arbitrary two-qubit observables, the architecture constrains the latent dynamics to structured correlation channels associated with physically meaningful spin interactions.

The observable embedding is defined as

$$\Phi : \mathbb{CP}^{63} \rightarrow \mathbb{R}^{27}.$$

The origin of this mapping can be understood from the quantum circuit Hilbert space.

The 6-qubit quantum circuit state lives in

$$(\mathbb{C}^2)^{\otimes 6},$$

whose dimension is

$$2^6 = 64.$$

A pure quantum state in a 64-dimensional complex Hilbert space is represented by the complex projective space

$$\mathbb{CP}^{63}.$$

The notation

$$\mathbb{CP}^{63}$$

denotes the space of all normalised complex vectors in

$$\mathbb{C}^{64},$$

modulo physically irrelevant global phase transformations

$$|\psi\rangle \sim e^{i\theta}|\psi\rangle.$$

Thus, two states differing only by a global phase represent the same physical quantum state.

The observable map

$$\Phi : \mathbb{CP}^{63} \rightarrow \mathbb{R}^{27}$$

projects the high-dimensional quantum state into a real-valued observable latent vector consisting of:

- 6 local Z expectation values,
- 6 local X expectation values,
- 5 ZZ correlations,
- 5 XX correlations,
- 5 YY correlations.

Therefore,

$$6 + 6 + 5 + 5 + 5 = 27,$$

and the resulting latent representation becomes

$$o_n = \Phi(|\Psi_n\rangle) \in \mathbb{R}^{27}.$$

Importantly, the latent vectors do not populate the ambient Euclidean space arbitrarily. Their structure is constrained by:

- quantum normalisation,
- operator algebra relations,
- symmetry constraints inherited from the underlying quantum state.

Consequently, the latent vectors occupy a structured observable manifold rather than an unconstrained Euclidean embedding.

This motivates the geometric interpretation of the latent space as a symmetry-organised correlator manifold.

Locally, the latent vectors transform under symmetry operations generated by the observable algebra. In this sense, the latent space behaves analogously to a coadjoint orbit

$$\mathcal{O}_\lambda \subset \mathfrak{g}^*,$$

where:

- $\mathfrak{g}^*$ denotes the dual Lie algebra space,

– λ labels the orbit under the group action.

Intuitively, this means the latent vectors are not merely static numerical features. Instead, they transform systematically under symmetry operations, similar to spin configurations in quantum lattice systems.

For this reason, the HVK latent manifold is described as an “SU(3)-variant latent space.” The term “variant” is important because the architecture does not explicitly construct the full $\mathfrak{su}(3)$ Lie algebra. Rather, the observable organisation and correlation structure behave analogously to a reduced symmetry representation with non-trivial transformation properties.

Thus, the latent space should be interpreted not as a standard Euclidean embedding, but as a structured observable manifold governed by quantum correlation geometry and symmetry-adapted operator dynamics.

6 Petubating Heisenberg Hamiltonian

6.1 Heisenberg Model

In condensed matter physics, the anisotropic (XYZ) Heisenberg Hamiltonian on a 1D chain of $n_{bonds} = 5$ bonds is:

$$H_{heisenberg} = \sum_{i=0}^4 (J_x^{[i]} X_i X_{i+1} + J_y^{[i]} Y_i Y_{i+1} + J_z^{[i]} Z_i X_{i+1})$$

where we can $J_x^{[i]}, J_y^{[i]}, J_z^{[i]} \in \mathbb{R}$ are the coupling strengths on the bond $(i, i+1)$. In the isotropic limit $J_x = J_y = J_z = J$ and the H reduces to the Heisenberg to an XXX model with SU(2) symmetry but the $U(1)$ breaks this generalization for the generic couplings makes the Hamiltonian Anisotropic

6.1.1 Learnable Coupling Strengths

In HVK, the coupling strengths

$$\{J_x^{(i)}, J_y^{(i)}, J_z^{(i)}\}$$

for each bond are learnable parameters initialised from a normal distribution:

$$J_x, J_y, J_z \in \mathbb{R}^{n_{bonds}}, \quad J_\alpha^{(i)} \sim \mathcal{N}(0, 0.01).$$

Because the coupling strengths are optimised jointly with the circuit weights, the Hamiltonian itself adapts to the image data, selecting the effective spin-spin interaction pattern that best captures the correlational structure of the patches.

6.2 Energy Loss and Ground-State Objective

Given the circuit output observables

$$\{ZZ, XX, YY\}$$

for patch n , the Heisenberg energy is

$$E_n(\theta, J) = \sum_{i=0}^4 \left[J_z^{(i)} \langle Z_i Z_{i+1} \rangle_n + J_x^{(i)} \langle X_i X_{i+1} \rangle_n + J_y^{(i)} \langle Y_i Y_{i+1} \rangle_n \right].$$

The energy loss minimised during training is the mean Heisenberg energy across all patches:

$$L_{\text{energy}}(\theta, J) = \langle E \rangle = \frac{1}{N} \sum_{n=1}^N E_n(\theta, J).$$

Minimising

$$L_{\text{energy}}$$

drives the circuit parameters toward quantum states that are ground states of the learned Hamiltonian. In contrast to minimising

$$|L_{\text{energy}}|,$$

which could drive the system toward any zero-energy eigenstate (including excited states with cancellation), directly minimising

$$L_{\text{energy}}$$

for negative J (antiferromagnetic coupling) favours the true ground state of the Hamiltonian, corresponding to highly entangled states. This inductive bias encourages the circuit to produce strongly correlated quantum features.

6.3 Positional Awareness of the Hamiltonian

The Hamiltonian is rendered spatially aware via two mechanisms. First, the per-qubit positional rotations

$$U_{\text{pos}}(\theta_{\text{pos},n})$$

depend on the patch-grid coordinate through the learned projection

$$W_{\text{pos}} \cdot e_n,$$

so distinct spatial positions produce different input states to the same Hamiltonian.

Second, since the circuit output state

$$|\Psi_n\rangle$$

depends on

$$(x_n, \theta_{\text{pos},n}),$$

the effective energy surface

$$E_n(\theta, J)$$

experienced by each patch is distinct even for identical MPS features.

The positional Heisenberg cost function is therefore

$$L_{\text{energy}}(n) = \sum_i \left[J_z^{(i)} \langle Z_i Z_{i+1} \rangle_{|\Psi_n(x_n, e_n)\rangle} + J_x^{(i)} \langle X_i X_{i+1} \rangle + J_y^{(i)} \langle Y_i Y_{i+1} \rangle \right],$$

where the dependence on the patch position n enters through the quantum state

$$|\Psi_n\rangle.$$

6.4 Total Training Objective

The complete loss is a weighted sum of the pixel-level reconstruction loss and the quantum energy loss:

$$L_{\text{total}} = L_{\text{rec}} + \lambda \cdot L_{\text{energy}},$$

with reconstruction loss

$$L_{\text{rec}} = \frac{1}{N} \sum_{n=1}^N \|D(o_n, e_n) - P_n\|_F^2,$$

where

$$D(\cdot)$$

denotes the classical decoder,

$$P_n$$

is the ground-truth image patch,

$$\|\cdot\|_F$$

is the Frobenius norm over pixel values, and

$$\lambda = 0.01$$

is the energy regularisation coefficient.

The decoder loss

$$L_{\text{rec}}$$

supervises the semantic reconstruction quality of the latent representation, while

$$L_{\text{energy}}$$

regularises the quantum circuit toward physically meaningful low-energy states.

7 Symmetry Analysis: From Continuous Correlator Symmetry to the Dihedral Group D_4

7.1 Overview of Symmetry Reduction

The HVK architecture exhibits a hierarchy of symmetry reductions as the image signal propagates through the quantum-classical pipeline. We trace this reduction from the continuous symmetry structure associated with the correlator manifold to the discrete dihedral group

$$D_4,$$

the symmetry group of the square patch grid.

This symmetry reduction is not an artefact of implementation but a structural consequence of introducing positional encoding and spatial patch geometry. The resulting symmetry-adapted latent structure enables the architecture to distinguish spatially localised image features while preserving geometric consistency across the image manifold.

7.2 Continuous Symmetry Structure of the Correlator Sector

The observable vector

$$o_n = \Phi(|\Psi_n\rangle) \in \mathbb{R}^{27}$$

is constructed from nearest-neighbour correlators of the form

$$\langle Z_i Z_{i+1} \rangle, \quad \langle X_i X_{i+1} \rangle, \quad \langle Y_i Y_{i+1} \rangle.$$

For a six-qubit circuit, these observables define the correlator sectors

$$\Xi_{ZZ} = (\langle Z_0 Z_1 \rangle, \dots, \langle Z_4 Z_5 \rangle),$$

$$\Xi_{XX} = (\langle X_0 X_1 \rangle, \dots, \langle X_4 X_5 \rangle),$$

$$\Xi_{YY} = (\langle Y_0 Y_1 \rangle, \dots, \langle Y_4 Y_5 \rangle),$$

which collectively define a structured observable manifold embedded in Euclidean space.

The underlying Heisenberg interaction

$$H = \sum_i \left(J_x^{(i)} X_i X_{i+1} + J_y^{(i)} Y_i Y_{i+1} + J_z^{(i)} Z_i Z_{i+1} \right)$$

possesses an approximate global $SU(2)$ symmetry in the isotropic limit

$$J_x = J_y = J_z.$$

This induces a symmetry organisation of the correlator manifold. Rather than identifying the latent space with an exact Lie algebra representation, we interpret the observable geometry as carrying an effective symmetry structure analogous to representation-theoretic sectors.

Motivated by this viewpoint, the latent space may be viewed locally as a symmetry-organised manifold whose coordinates transform covariantly under continuous rotations of the correlator sector.

7.3 Symmetry Reduction by Positional Encoding

The positional modulation operator

$$U_{\text{pos}}(\theta_{\text{pos},n})$$

applies distinct single-qubit rotations to each qubit. These rotations belong to

$$U(1)^{\otimes 6} \subset SU(2)^{\otimes 6}.$$

Because the positional angles differ across qubits, permutation symmetry between qubit sites is explicitly broken. Consequently, the continuous symmetry of the isotropic correlator sector reduces to the subgroup compatible with the positional configuration.

The remaining symmetry is approximately described by diagonal phase rotations associated with the Cartan sector of the underlying continuous symmetry algebra. Operationally, this means that positional encoding selects preferred directions in latent space, allowing distinct spatial regions of the image to generate distinguishable quantum states.

7.4 Discretisation by the Patch Grid Geometry

A second symmetry reduction arises from the geometry of the image patch grid itself. The 4×4 patch arrangement inherits the symmetries of the square, whose symmetry group is the dihedral group

$$D_4 = \langle r, s \mid r^4 = s^2 = (sr)^2 = e \rangle.$$

The group D_4 contains eight elements corresponding to

$$0^\circ, 90^\circ, 180^\circ, 270^\circ$$

rotations together with four reflections.

Under a geometric transformation

$$g \in D_4,$$

the observable vectors transform approximately as

$$o_{g(i,j)} \approx \Phi_g o_{(i,j)},$$

where

$$\Phi_g \in O(27)$$

is an orthogonal transformation describing the induced permutation and reflection of observable components.

Consequently, the latent representation may be organised into approximately D_4 -equivariant sectors.

7.5 Irreducible Representation Structure of the Latent Space

The group D_4 possesses five irreducible representations:

$$A_1, A_2, B_1, B_2, E.$$

The observable space may therefore be interpreted as decomposing into symmetry-sensitive feature sectors:

$$V_{\text{obs}} \approx m_{A_1} A_1 \oplus m_{A_2} A_2 \oplus m_{B_1} B_1 \oplus m_{B_2} B_2 \oplus m_E E,$$

where the multiplicities

$$m_\rho$$

depend on the variational ansatz and learned Hamiltonian couplings.

Features associated with the trivial representation

$$A_1$$

encode approximately rotation-invariant patch statistics, while sectors transforming under

$$E$$

capture directional and orientation-sensitive information such as edges and textures.

7.6 Architectural Role of Symmetry Reduction

The symmetry reduction from the continuous correlator sector to the discrete group

$$D_4$$

is essential for high-resolution image reconstruction.

A fully continuous symmetry-invariant latent space would map many spatially distinct image patches onto equivalent latent representations, reducing the decoder’s ability to distinguish orientations and local geometric structure.

The reduction to D_4 -equivariant sectors preserves precisely the geometric symmetries naturally exhibited by the square patch grid while breaking excessive degeneracies of the continuous latent manifold.

As a result, the decoder can distinguish:

horizontal vs vertical structure,

rotationally symmetric vs asymmetric features,

and

reflection-symmetric vs antisymmetric textures.

This mechanism is conceptually analogous to equivariant convolutional architectures, although here the symmetry structure emerges from the interaction between Hamiltonian dynamics, correlator geometry, and positional modulation within the quantum latent space.

References

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023.